

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

HYPERBOLA

DIBY (DO IT BY YOURSELF)

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HYPERBOLA



DIBY-01

The eccentricity of the hyperbola whose length of the latusrectum is equal to 8 and the length of its conjugate axis is equal to half of the distance between its foci, is

[JEE Mains-2017]

- (A) $\frac{4}{3}$ (B) $\frac{4}{\sqrt{3}}$ (C) $\frac{2}{\sqrt{3}}$ (D) $\sqrt{3}$

[Ans. : (C)]

DIBY-02

Let $0 < \theta < \frac{\pi}{2}$. If the eccentricity of the hyperbola $\frac{x^2}{\cos^2 \theta} - \frac{y^2}{\sin^2 \theta} = 1$ is greater than 2, then the length of its latus rectum lies in the interval

[JEE Mains-2019]

- (A) $\left(1, \frac{3}{2}\right]$ (B) $(3, \infty)$ (C) $\left(\frac{3}{2}, 2\right]$ (D) $(2, 3]$

[Ans. : (B)]

DIBY-03

A hyperbola, having the transverse axis of length $2\sin\theta$, is confocal with the ellipse $3x^2 + 4y^2 = 12$. Then, its equation is

[JEE-2007, 3M]

- (A) $x^2 \operatorname{cosec}^2 \theta - y^2 \sec^2 \theta = 1$ (B) $x^2 \sec^2 \theta - y^2 \operatorname{cosec}^2 \theta = 1$
(C) $x^2 \sin^2 \theta - y^2 \cos^2 \theta = 1$ (D) $x^2 \cos^2 \theta - y^2 \sin^2 \theta = 1$

[Ans. : (A)]

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DIBY-04

Let the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be reciprocal to that of the ellipse $x^2 + 4y^2 = 4$. If the hyperbola passes through a focus of the ellipse, then

[JEE-2011]

- (A) the equation of the hyperbola is $\frac{x^2}{3} - \frac{y^2}{2} = 1$
- (B) a focus of the hyperbola is $(2, 0)$
- (C) the eccentricity of the hyperbola is $\sqrt{\frac{5}{3}}$
- (D) the equation of the hyperbola is $x^2 - 3y^2 = 3$

[Ans. : (B), (D)]

DIBY-05

For hyperbola $\frac{x^2}{\cos^2 \alpha} - \frac{y^2}{\sin^2 \alpha} = 1$, which of the following remains constant with change in ' α '?

[JEE-2003, 1M]

- | | |
|---------------------------|-----------------------|
| (A) Abscissae of vertices | (B) Abscissae of foci |
| (C) Eccentricity | (D) Directrix |

[Ans. : (B)]

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DIBY-06

If $5x + 9 = 0$ is the directrix of the hyperbola $16x^2 - 9y^2 = 144$, then its corresponding focus is **[JEE Mains-2019]**

- (A) $\left(-\frac{5}{3}, 0\right)$ (B) $(-5, 0)$ (C) $\left(\frac{5}{3}, 0\right)$ (D) $(5, 0)$

[Ans. : (B)]

DIBY-07

If the vertices of a hyperbola be at $(-2, 0)$ and $(2, 0)$ and one of its foci be at $(-3, 0)$, then which one of the following points does not lie on this hyperbola? **[JEE Mains-2019]**

- (A) $(2\sqrt{6}, 5)$ (B) $(6, 5\sqrt{2})$ (C) $(4, \sqrt{15})$ (D) $(-6, 2\sqrt{10})$

[Ans. : (B)]

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DIBY-08

If a directrix of a hyperbola centred at the origin and passing through the point $(4, -2\sqrt{3})$ is $5x = 4\sqrt{5}$ and its eccentricity is e , then

(A) $4e^4 - 12e^2 - 27 = 0$

(B) $4e^4 - 24e^2 + 27 = 0$

(C) $4e^4 + 8e^2 - 35 = 0$

(D) $4e^4 - 24e^2 + 35 = 0$

[Ans. : (D)]

DIBY-09

Find the locus of the foot of perpendicular from the centre upon any normal to the

hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

[Ans. : $(x^2 + y^2)^2 (a^2 y^2 - b^2 x^2) = (a^2 + b^2)^2 x^2 y^2$]

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DIBY-10

Show that the chord joining two points, whose sum of parametric angle is $\frac{\pi}{2}$ passes through a fixed point for hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$.

DIBY-11

If S_1 and S_2 are the foci of the hyperbola, whose transverse axis length is 4 units and conjugate axis length is 6 units and S_3 and S_4 are the foci of the conjugate hyperbola, then find the area of the quadrilateral $S_1 S_3 S_2 S_4$.

[Ans. : 26]

DIBY-12

Let $P(a \sec \theta, b \tan \theta)$ and $Q(a \sec \phi, b \tan \phi)$ where $\theta + \phi = \frac{\pi}{2}$, be two points on $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If (h, k) is the point of intersection of the normals at P and Q , then K is equal to

- (A) $\frac{a^2+b^2}{a}$ (B) $-\left(\frac{a^2+b^2}{a}\right)$ (C) $\frac{a^2+b^2}{b}$ (D) $-\left(\frac{a^2+b^2}{b}\right)$

[Ans. : (A)]

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DIBY-13

An ellipse has eccentricity $\frac{1}{2}$ and one focus at the point $P\left(\frac{1}{2}, 1\right)$. Its one directrix is the common tangent, nearer to the point P , to $x^2 + y^2 = 1$ and the hyperbola $x^2 - y^2 = 1$. The equation of the ellipse is

(A) $\frac{\left(x - \frac{1}{3}\right)^2}{\frac{1}{9}} + \frac{(y-1)^2}{\frac{1}{12}} = 1$

(B) $\frac{\left(x - \frac{1}{3}\right)^2}{\frac{1}{8}} + \frac{(y-1)^2}{\frac{1}{12}} = 1$

(C) $\frac{\left(x - \frac{1}{3}\right)^2}{\frac{1}{9}} + \frac{(y-1)^2}{\frac{1}{8}} = 1$

(D) None of these

[Ans. : (D)]

DIBY-14

If a quadrilateral formed by four tangents to the ellipse $3x^2 + 4y^2 = 12$ is a square, then

- (A) The vertices of the square lie on $y = \pm x$
- (B) The vertices of the square lie on $x^2 + y^2 = 7^2$
- (C) The area of all such squares is constant
- (D) Only two such squares are possible

[Ans. : (B), (C)]

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DIBY-15

The locus of a point whose chord of contact touches the circle described on the straight line joining the foci of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ as diameter is/has

(A) Hyperbola

(B) Ellipse

(C) $e = \frac{\sqrt{a^4 - b^4}}{a^2}$

(D) $e = \frac{\sqrt{a^4 + b^4}}{a^2}$

[Ans. : (B), (C)]

DIBY-16

An ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through the vertices of the hyperbola $H: \frac{x^2}{49} - \frac{y^2}{64} = -1$. Let the major and minor axes of the ellipse E coincide with the transverse and conjugate axes of the hyperbola H , respectively. Let the product of the eccentricities of E and H be $\frac{1}{2}$. If ℓ is the length of the latus rectum of the ellipse E , then the value of 113ℓ is equal to

[JEE Mains-2022]

[Ans. : 1152]

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DIBY-17

The locus of the centroid of the triangle formed by any point P on this hyperbola

$16x^2 - 9y^2 + 32x + 36y - 164 = 0$, and its foci is: [JEE Mains-2021]

- (A) $16x^2 - 9y^2 + 32x + 36y - 144 = 0$ (B) $9x^2 - 16y^2 + 36x + 32y - 144 = 0$
(C) $16x^2 - 9y^2 + 32x + 36y - 36 = 0$ (D) $9x^2 - 16y^2 + 36x + 32y - 36 = 0$

[Ans. : (C)]

DIBY-18

Let $a > 0, b > 0$. Let e and ℓ respectively be the eccentricity and length of the latus rectum of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Let e' and ℓ' respectively be the eccentricity and length of the latus rectum of its conjugate hyperbola. If $e^2 = \frac{11}{14}\ell$ and $(e')^2 = \frac{11}{8}\ell'$, then the value of $77a + 44b$ is equal to: [JEE Mains-2022]

- (A) 100 (B) 110 (C) 120 (D) 130

[Ans. : (D)]

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DIBY-19

Let the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be $\frac{5}{4}$. If the equation of the normal at the point $\left(\frac{8}{\sqrt{5}}, \frac{12}{5}\right)$ on the hyperbola is $8\sqrt{5}x + \beta y = \lambda$, then $\lambda - \beta$ is equal to **[JEE Mains-2022]**

[Ans. : 20]

DIBY-20

Let $H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, a > 0, b > 0$, be a hyperbola such that the sum of lengths of the transverse and the conjugate axes is $4(2\sqrt{2} + \sqrt{14})$. If the eccentricity H is $\frac{\sqrt{11}}{2}$, then the value of $a^2 + b^2$ is equal to. **[JEE Mains-2022]**

[Ans. : 88]

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DIBY-21

Let the equation of two diameters of a circle $x^2 + y^2 - 2x + 2fy + 1 = 0$ be $2px - y = 1$ and $2x + py = 4p$. Then the slope $m \in (0, \infty)$ of the tangent to the hyperbola $3x^2 + y^2 = 3$ passing through the centre of the circle is equal to

[JEE Mains-2022]

[Ans. : 2]

DIBY-22

Let the hyperbola $H : \frac{x^2}{a^2} - y^2 = 1$ and the ellipse $E : 3x^2 + 4y^2 = 12$ be such that the length of latus rectum of H is equal to the length of latus rectum of E . If e_H and e_E are the eccentricities of H and E respectively, then the value of $12(e_H^2 + e_E^2)$ is equal to

[JEE Mains-2022]

[Ans. : 42]

DIBY-23

Let the sum of the focal to distances of the point $P(4, 3)$ on the hyperbola $H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be $8\sqrt{\frac{5}{3}}$. If for H , the length of the latus rectum is l and the product of the focal distances of the point P is m , then $9l^2 + 6m$ is equal [JEE Mains-2025]

- (A) 184 (B) 186 (C) 185 (D) 187

[Ans.: (C)]

DIBY-24

Let e_1 and e_2 be the eccentricities of the ellipse $\frac{x^2}{b^2} + \frac{y^2}{25} = 1$ and the hyperbola $\frac{x^2}{16} - \frac{y^2}{b^2} = 1$, respectively. If $b < 5$ and $e_1 e_2 = 1$, then the eccentricity of the ellipse having its axes along the coordinate axes and passing through all four foci (two of the ellipse and two of the hyperbola) is:

- (A) $\frac{4}{5}$ (B) $\frac{3}{5}$ (C) $\frac{\sqrt{7}}{4}$ (D) $\frac{\sqrt{3}}{2}$

[Ans.: (B)]

DIBY-25

Let $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b$ and $H: \frac{x^2}{A^2} - \frac{y^2}{B^2} = 1$. Let the distance between the foci of E and the foci of H be $2\sqrt{3}$. If $a - A = 2$, and the ratio of the eccentricities of E and H is $\frac{1}{3}$, then the sum of the lengths of their latus rectums is equal to **[JEE Mains-2025]**

- (A) 9 (B) 7 (C) 10 (D) 8

[Ans. : (D)]

DIBY-26

Let the circle C touch the line $x - y + 1 = 0$, have the centre on the positive x -axis, and cut off a chord of length $\frac{4}{\sqrt{13}}$ along the line $-3x + 2y = 1$. Let H be the hyperbola $\frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = 1$, whose one of the foci is the centre of C and the length of the transverse axis is the diameter of C . Then, $2\alpha^2 + 3\beta^2$ is equal to _____. **[JEE Mains-2025]**

[Ans. : 19]

DIBY-27

Let $H_1: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $H_2: -\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$ be two hyperbolas having length of latus rectums $15\sqrt{2}$ and $12\sqrt{5}$ respectively. Let their eccentricities be $e_1 = \sqrt{\frac{5}{2}}$ and e_2 respectively. If the product of the lengths of their transverse axes is $100\sqrt{10}$, then $25e_2^2$ is equal to ____.

[JEE Mains-2025]

[Ans. : 55]

DIBY-28

Let $0 < \theta < \frac{\pi}{2}$. If the eccentricity of the hyperbola $\frac{x^2}{\cos^2 \theta} - \frac{y^2}{\sin^2 \theta} = 1$ is greater than 2, then the length of its latus rectum lies in the interval:

[JEE Mains-2019]

- (A) $(3, \infty)$ (B) $\left(\frac{3}{2}, 2\right]$ (C) $(2, 3]$ (D) $\left(1, \frac{3}{2}\right]$

[Ans. : (A)]

DIBY-29

Let P be a point on the hyperbola $H: \frac{x^2}{9} - \frac{y^2}{4} = 1$, in the first quadrant such that the area of triangle formed by P and the two foci of H is $2\sqrt{13}$. Then, the square of the distance of P from the origin is [JEE Mains-2024]

- (A) 18 (B) 26 (C) 22 (D) 20

[Ans. : (C)]

DIBY-30

If the foci of a hyperbola are same as that of the ellipse $\frac{x^2}{9} + \frac{y^2}{25} = 1$ and the eccentricity of the hyperbola is $\frac{15}{8}$ times the eccentricity of the ellipse, then the smaller focal distance of the point $\left(\sqrt{2}, \frac{14}{3}\sqrt{\frac{2}{5}}\right)$ on the hyperbola, is equal to [JEE Mains-2024]

- (A) $7\sqrt{\frac{2}{5}} - \frac{8}{3}$ (B) $14\sqrt{\frac{2}{5}} - \frac{4}{3}$ (C) $14\sqrt{\frac{2}{5}} - \frac{16}{3}$ (D) $7\sqrt{\frac{2}{5}} + \frac{8}{3}$

[Ans. : (A)]

DIBY-31

Let the foci and length of the latus rectum of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$ be $(\pm 5, 0)$ and $\sqrt{50}$, respectively. Then, the square of the eccentricity of the hyperbola $\frac{x^2}{b^2} - \frac{y^2}{a^2 b^2} = 1$ equals

[JEE Mains-2024]

[Ans. : 51]

DIBY-32

For $0 < \theta < \pi/2$, if the eccentricity of the hyperbola $x^2 - y^2 \operatorname{cosec}^2 \theta = 5$ is $\sqrt{7}$ times eccentricity of the ellipse $x^2 \operatorname{cosec}^2 \theta + y^2 = 5$, then the value of θ is: [JEE Mains-2024]

(A) $\frac{\pi}{6}$

(B) $\frac{5\pi}{12}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{4}$

[Ans. : (C)]



*It is never about **end result**
It's always about the **Journey***

THANKYOU

#future **IITians**